

Vertical dilations

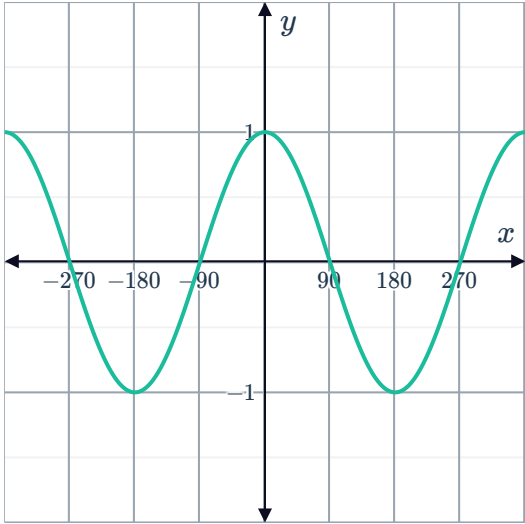


1

a

θ	0	60	90	120	180	240	270	300	360
$\cos \theta$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



c 1

d -1

e $[-4, 4]$

2

a $(-\infty, \infty)$

b $[-1, 1]$

3

a $(-\infty, \infty)$

b $[-2, 2]$

4 8

5

a $y = 2 \sin x$

b $y = -\sin x$

c $y = 3 \cos x$

d $y = -1.5 \cos x$

e $y = 3 \sin x$

f $y = 2.5 \sin x$

6 270

7

a 8

b 0

c -8

8 5

9 $-2 \sin x$

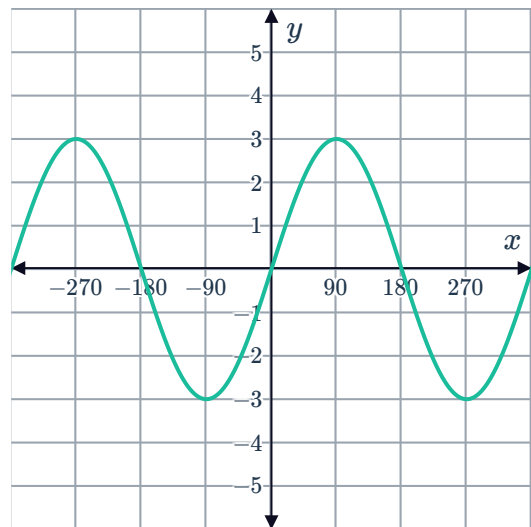


10

a

i $A = 3$

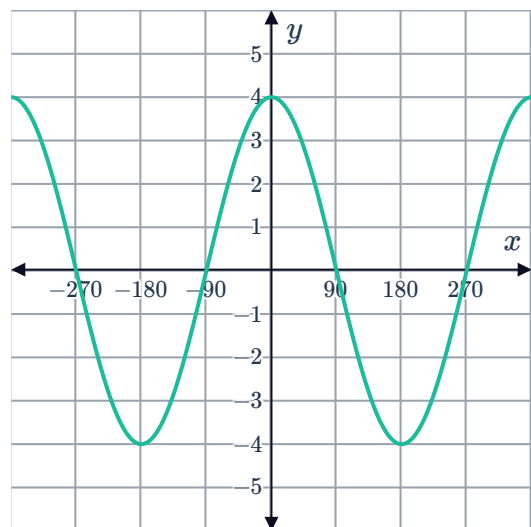
ii



b

i $A = 4$

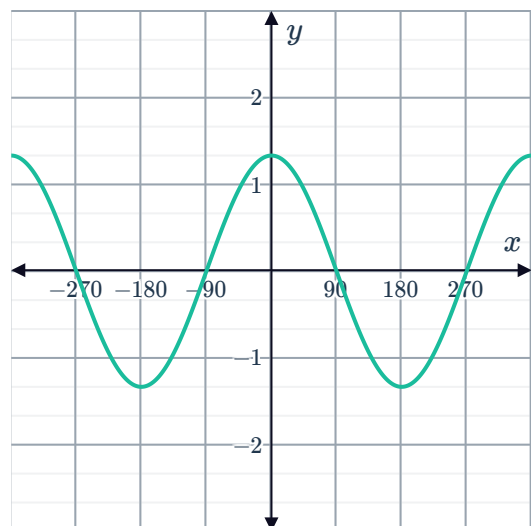
ii



c

i $A = \frac{4}{3}$

ii

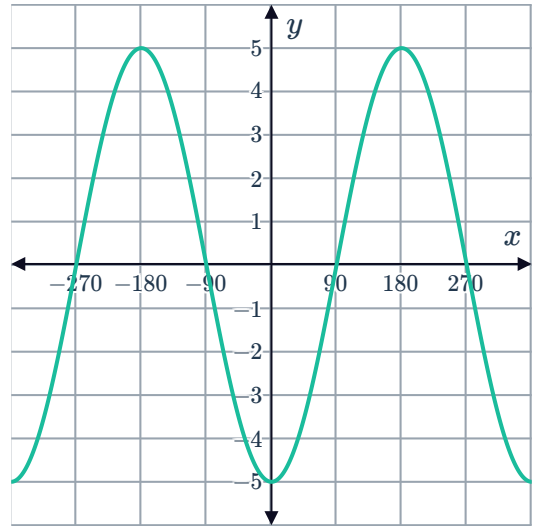


d

i $A = 5$



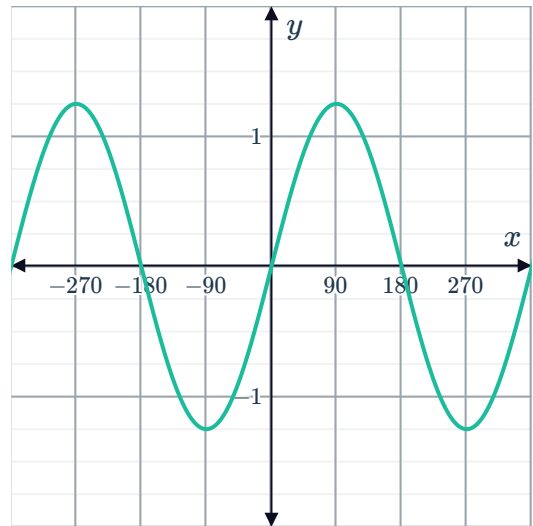
ii



e

i $\frac{5}{4}$

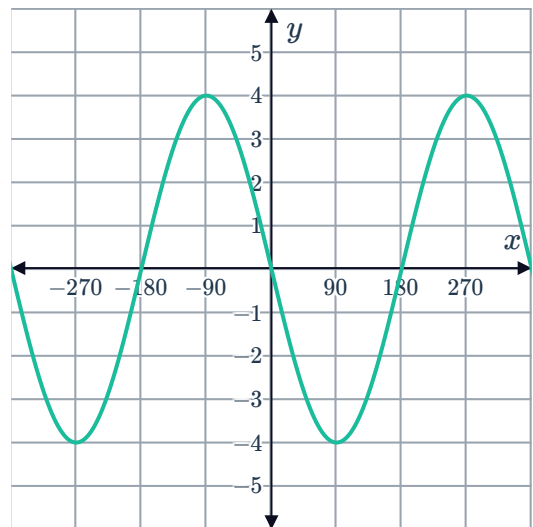
ii



f

i $A = 4$

ii



b -3

c 3

d A vertical dilation by a factor of 3 and reflection about the x -axis.



Vertical translations

12 Translated 2 units downwards.

13 Translated 2 units upwards.

14 Translated 4 units upwards

15

a $y = \sin x + 4$

c $y = \sin x - 4$

b $y = \sin x - 2$

d $y = \sin x + 1$

16

a $y = \cos x + 3$

c $y = \cos x - 3$

b $y = \cos x - 4$

d $y = \cos x + 2$

17 Reflection about the x -axis, then translation 3 units up.

18 $y = 2$

19

a Translated 4 units upwards.

c 5

b 360°

d 3

20

a

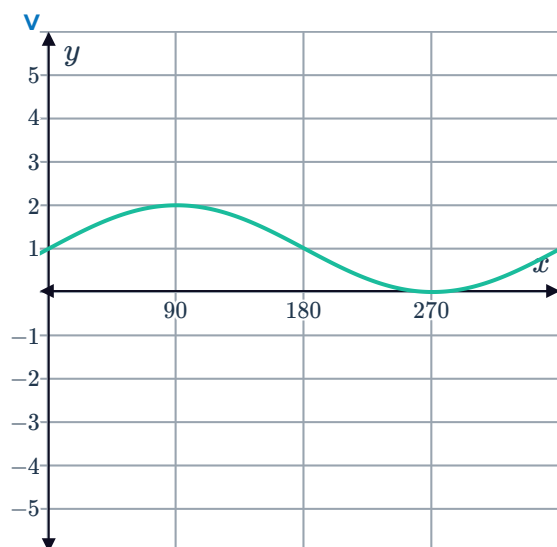
i 360

ii 1



iii 2

iv 0



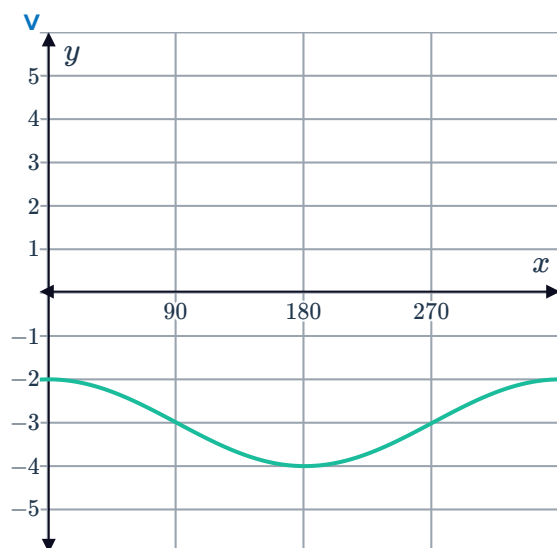
b

i 360

ii 1

iii -2

iv -4



c

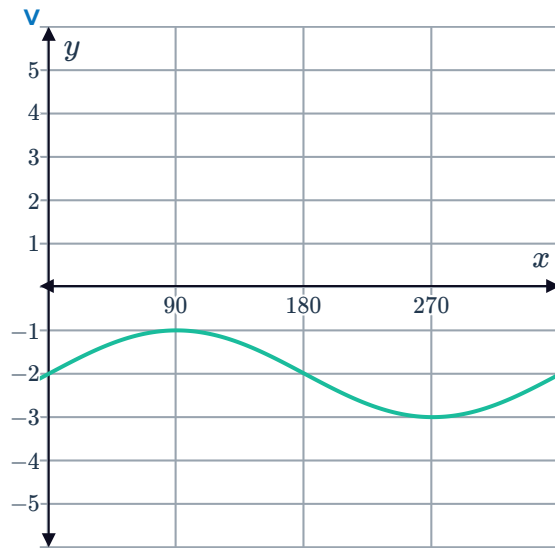
i 360

ii 1



iii -1

iv -3



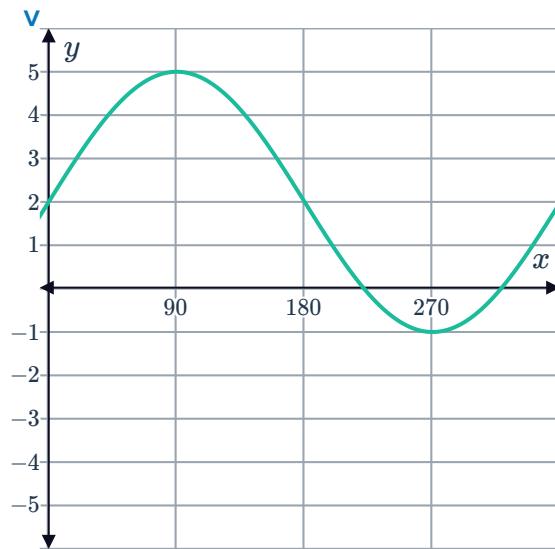
d

i 360

ii 3

iii 5

iv -1



e

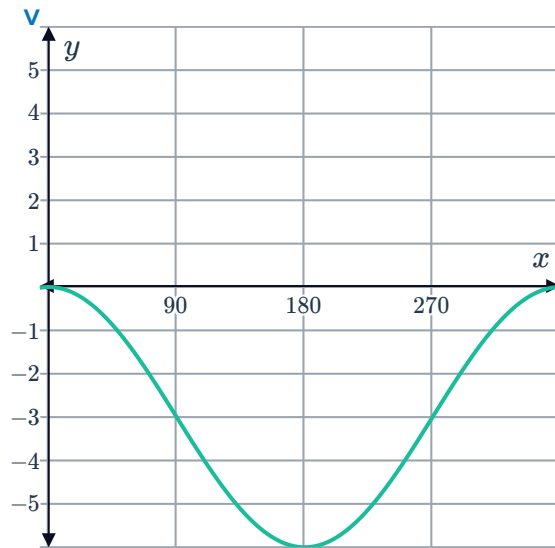
i 360

ii 3



iii 0

iv -6



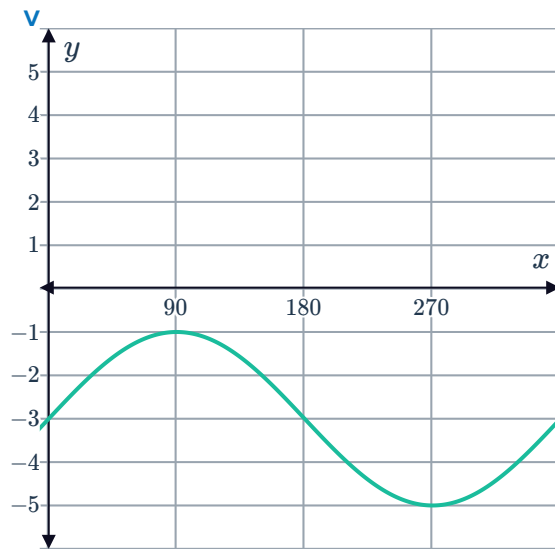
f

i 360

ii 2

iii -1

iv -5



21



a

i 3

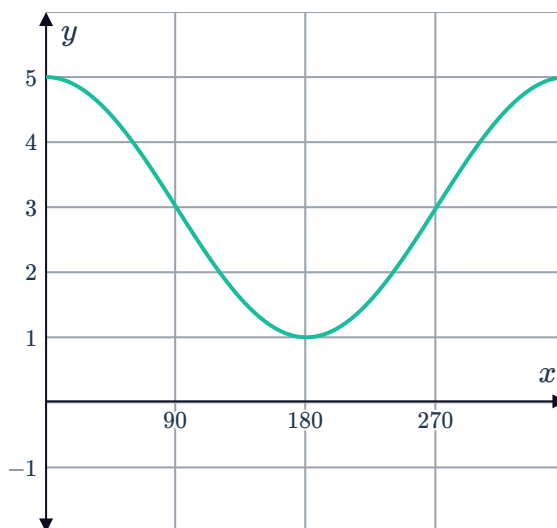
ii 2

iii 5

iv 1

v A vertical dilation by a factor of 2, then translated 3 units upwards.

vi



b

i -3

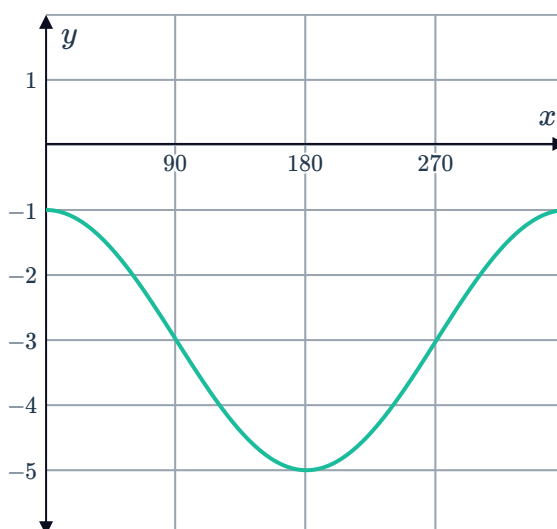
ii -1

iii -5

iv 2

v A vertical dilation by a factor of 2, then translated 3 units downwards.

vi



c

i 2

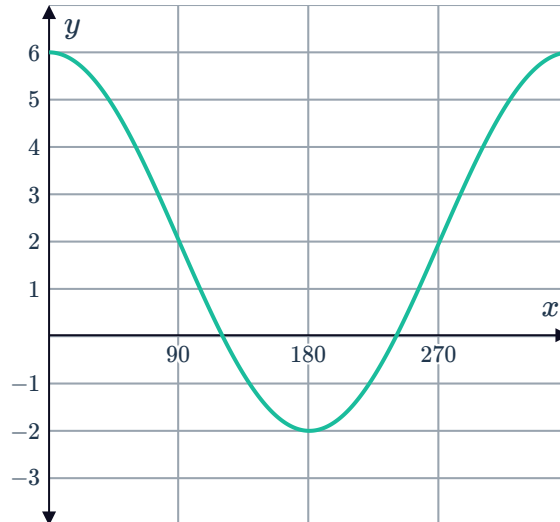
ii 6

iii -2

iv 4

v A vertical dilation by a factor of 4, a reflection about the x -axis and then translated 2 units upwards.

vi



d

i 2

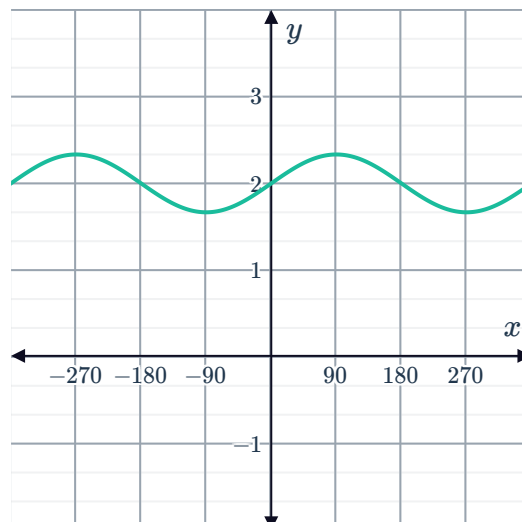
ii $\frac{7}{3}$

iii $\frac{5}{3}$

iv $\frac{1}{3}$

v A vertical dilation by a factor of $\frac{1}{3}$, then translated 2 units upwards.

vi



22 a $y = \cos x + 9$



b 10

23

a

i 5

ii -1

b

i 7

ii 1

c

i 2

ii 1

d

i 2

ii $\frac{1}{2}$

24 $2 \sin x + 2$

25 $8 \cos x - 2$